COMP9001
Lecture 4
Introduction to Programming Containers

Presenter: Dr Armin Chitizadeh

Version 0.1

THE UNIVERSITY OF SYDNEY

Inspired by Hazem El-Alfy 2024 slides

CRICOS 00026A TEQSA PRV12057

We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

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Consultations

- After Lecture
- Monday 12pm – 1pm Online
- Thursdays 5:30pm to 6:30pm (Hybrid)
- And more in the future!

Details on ed,
(There will be a page about it to be better organised)

✏️ Assessments

✓ Homework 1 (5 Seconds of Summer)

Due: Sun March 22nd, 11:59pm

Please remember about our limitations on answering the forum

Expect 2 business days,
only during working hours,
no weekend pr public-holidays answering

Mid-semester Review Intense Workshop

- Friday of mid-semester break (More info to be announced soon!)
  - Details might change

Assessment Schedule

Assessment Schedule

Task	Description	Release Week	Due Week	Weighting
Homework 1	Coding assessment to test student understanding of topics covered.	Week 1	Week 3	5%
Assignment 1	Coding assessment to test student understanding of topics covered. Focused on introductory topics.	Week 5	Week 7	15%
Assignment 2	Coding assessment to test student understanding of topics covered. Focused on more challenging topics in the unit.	Week 8	Week 11	25%
In-Lab-Checkpoint	At the end of each lab, students are asked to do a simple quiz. The primary purpose of these assessment is to ensure that students understand the topic thoroughly before concluding the tutorial. The highest 8 marks out of a total of 12 will be used to calculate the mark.	N/A	Weekly post-lab	5%

“Yay” moment!

Photo by Jonathan Borba on Unsplash

”Yay” moment!

https://www.menti.com/alxohsrkkw1g

Where are we in the unit?

Week 1 Intro

Week 2 Variables

Week 3 Conditionals

Week 4 containers

Week 5 Flow/Functions

Week 6 Iterations

Week 7 Classes/Objects

Week 8 File I/O

Week 9 Adv Flow

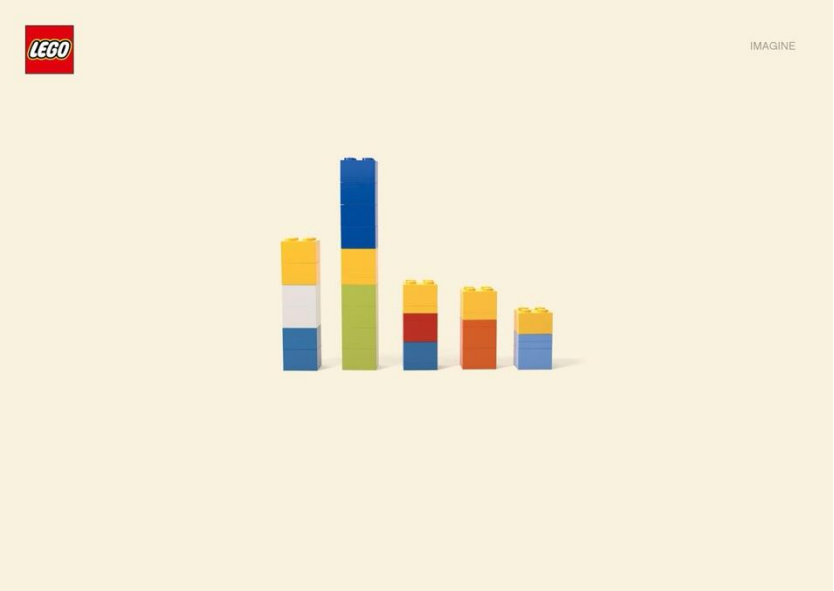
Week 10 Testing

Week 11 Recursion

Week 12 Multi Arrays

Week 13 Revision

Final Project



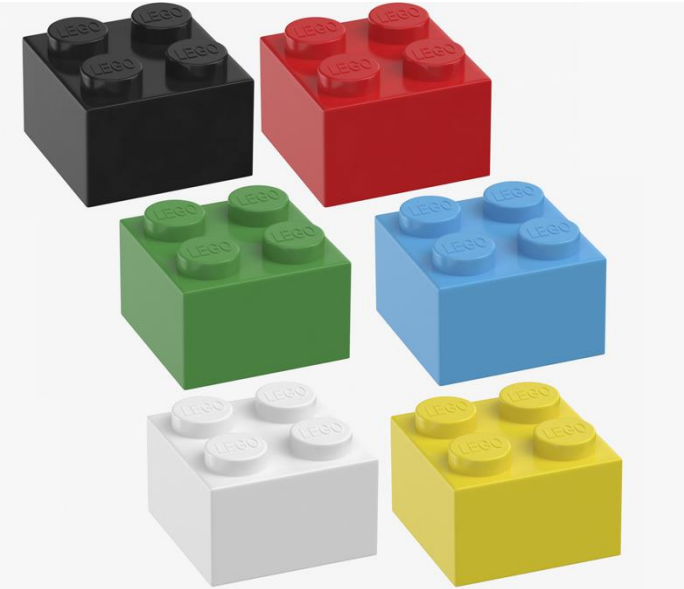
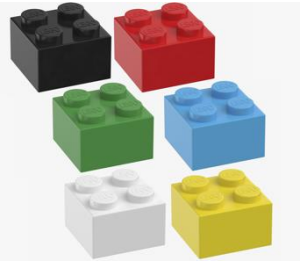


Image credit:
https://www.adsoftheworld.com/media/print/lego_the_simpsons
https://www.turbosquid.com/3d-models/lego-bricks-2x2-3d-model-1491947

Data Types

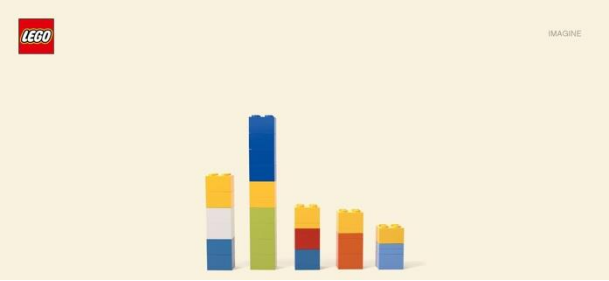
<u>Basic</u> (Single Value)

- Integers
- Floating Points
- Boolean
- Characters



<u>Containers</u> (Multiple Values)

- Strings
- Lists
- Tuples
- Dictionaries



Esfandiar's Seven Labours

Artist: Skhokoufeh Hassanzarei
Link: https://artrepreneur.com/p/2YSHw7uu33PQjdWj7/art/ctmb96YKqFYW6iT7W

Esfandiar's Seven Labours

- How do we store these:
    1. Two monstrous wolves
    2. Two man eating lions
    3. Dragon
    4. Evil Magic
    5. Simurgh
    6. Storm
    7. Desert

esfandiar_seven_labour_bad.py

```python
# Storing Esfandiar's Seven Labours
# Not the best way

battle1 = "wolves"
battle2 = "lions"
battle3 = "dragon"
battle4 = "evil_magic"
battle5 = "simurgh"
battle6 = "storm"
battle7 = "desert"

print(battle1, battle2, battle3,
    battle4, battle5, battle6, battle7)
```

list

List – What and Why?

- A *data structure* is a way to organize data that we wish to process with a computer program.

- A list is a *contiguous* block of memory containing multiple values not all of the same type.

- List solves the problem of having many variables.

Python List

- A multi-element list is defined as comma separated objects within the square brackets [and]

- Elements of the same type = homogenous list
  Elements of different types = heterogenous list

- Python's list is <u>NOT</u> the same as arrays, i.e.

  arrays and lists are **not** the same in Python!

Esfanidar's Seven Labour List

esfandiar_seven_labour.py

```python
# Storing Esfandiar's Seven Labours
# A better way
battle_list = ["wolves", "lions", "dragon",
                        "evil_magic", "simurgh", "storm",
                        "desert"]

print(battle_list)
```

Initialisation

- Many ways to initialize lists.
- Can you spot examples of each initialization below?

```
# Initialize lists
empty_ls = []
tutors = ["Avinash", "Joshua", "Luis"]
x = [5, 10, 15, 20, 25]
help_ls = list("help")
```

Python List

- How to access them individually?

```
battle_list[0]
```

- How to measure length of a list?

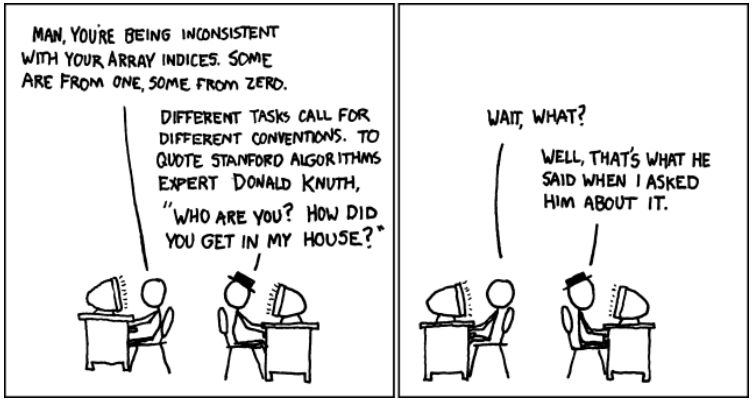
```
len(battle_list)
```

- How to update them?

```
battle_list[1] = "beast_akvan"
```

Zero-based Indexing and One-based Indexing

- There are programming languages that use one-based indexing. The first element starts at one instead of zero.
- Consequences:
    - Index-1: Index of last element is the **<u>same</u>** as total number of elements.
    - Index-0: Index of last element is **<u>one less than</u>** the total number of elements.



- Python is Zero-based

Rostam's Seven Labours

"I want to record everything as I go through"

- Rostam

- How do we store these once they happen:
    1. Lion
    2. Thirst
    3. Dragon
    4. Temptress
    5. Beasts
    6. Rescue the King
    7. White Beast

Artist:
Saeed Ramez
https://au.pinterest.com/pin/412360909638705440/

Playing with Lists in Python!

List Manipulation

Method	Description
`append(x)`	Adds an element **`x`** to the end of the list.
`count(x)`	Returns the number of times element **`x`** appears in the list.
`index(x)`	Returns the index of the first occurrence of element **`x`** in the list.
`insert(index, x)`	Inserts an element **`x`** at a given **`index`**. Note that the first element in the list has index **`0`**.
`pop(index)`	Removes the element at the given index and return it. The parameter index is optional. If it is not specified, **`list.pop()`** removes and returns the last element in the list.
`remove(x)`	Removes the first occurrence of element **`x`** from the list.
`reverse(x)`	Reverses the elements in the list.
`sort()`	Sorts the elements in the list in an ascending order.

Liang, Y, et al. *Introduction to Python Programming and Data Structures, Global Edition*, Pearson Education, Limited, 2022. *ProQuest Ebook Central,*

List Manipulation

- Rostam Python Demo!

list1 + list2

```
[1,2,3] + [4,5,6] =  [1,2,3,4,5,6]
```

Slicing in Python

Slicing

- The slicing operator returns a slice of the given list using the syntax **`list[start:end]`**
  - The `start` index is the start of the slice (**inclusive**)
  - The `end` index is the end of the slice (**exclusive**)
- For `alphabet = ['a', 'b', 'c', 'd', 'e', 'f']` try:
  - `alphabet[2:5]`
  - `alphabet[2:]`
  - `alphabet[:5]`
  - `alphabet[-1]`
  - `alphabet[-3]`
  - `alphabet[-7]`

Practice Slicing on Ed Lesson!

Examples by Joshua Blair and Ehab Ahmed

Slicing with step

- The slicing operator returns a slice of the given list using the syntax **`list[start:end:step]`**
  - The `start` index is the start of the slice (**inclusive**)
  - The `end` index is the end of the slice (**exclusive**)
  - The `step` is interval between elements to include in a slice
- For `alphabet = ['a', 'b', 'c', 'd', 'e', 'f']` try:
  - `alphabet[0:6:2]`
  - `alphabet[::2]`
  - `alphabet[::3]`
  - `alphabet[::-1]`

Two Dimensional lists

- Lists can be two dimensional!

Two Dimensional Lists

```
game = [['x', '-', 'o'], ['-', 'x', 'o'], ['-', '-', 'o']]
```

```
game = [
    ['x', '-', 'o'],
    ['-', 'x', 'o'],
    ['-', '-', 'o']
]
```

Accessing

- Try:
    - `print(game)`
    - `print(game[1])`
    - `print(game[1][2])`

Tuples

Tuples

- You can use tuples for storing a fixed list of elements
- Tuples are like lists, but their elements are fixed
- Once tuple is created, you cannot add , delete, or replace any element **accidentally**

- A multi-element tuple is defined as comma separated objects within the parentheses `(` and `)`
- You can do the same operations you did on `list` on `tuples` too?

Try the ed lesson yourself!

Mutable and Immutable

- List and tuples have common operations:
  - Length using `len(empty_ls)`
  - Accessing `i`th item of s from origin `0` e.g. `tutors[1]`
  - Slices from index `i` to `j-1` e.g. `tutors[1:2]`

- Lists are **mutable**. Tuples are **immutable**.
  - You can modify lists' elements after creation.
  - Like string objects, once you create a tuple, you cannot modify its elements.
  - Tuples are used to pass multiple values out a function to reduce the risk of modifying the function's return values.

Deep Dive Into How Lists and Tuples Are Implemented in Python

- You take the **blue pill** – the story ends, you wake up in your bed and believe whatever you want to believe.

- You take the **red pill** – you stay in Wonderland, and I show you how deep the rabbit hole goes.

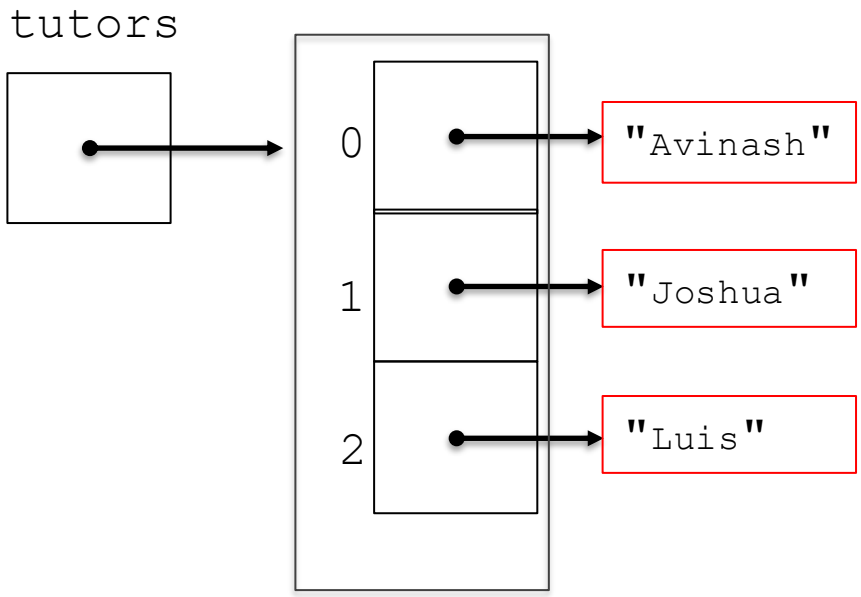
Vote:
Deep Dive Into How Lists and Tuples Are Implemented in Python

- You take the **blue pill** – the story ends, you wake up in your bed and believe whatever you want to believe.

- You take the **red pill** – you stay in Wonderland, and I show you how deep the rabbit hole goes

Object Level Trace Diagram

```
1 # Initialize lists
2 empty_ls = []
3 tutors = ["Avinash", "Joshua", "Luis"]
4 ls_one_element = [5]
```



String, List, Tuples

```python
greet = 'hello'
print(greet[0])
print(greet[1])
print(greet[2])
print(greet[3])
print(greet[4])
# End program
```

String, List, Tuples

- Can you convert a string to list and tuple?

- Does this work:

```
"Hello"[0]
```

Menti!

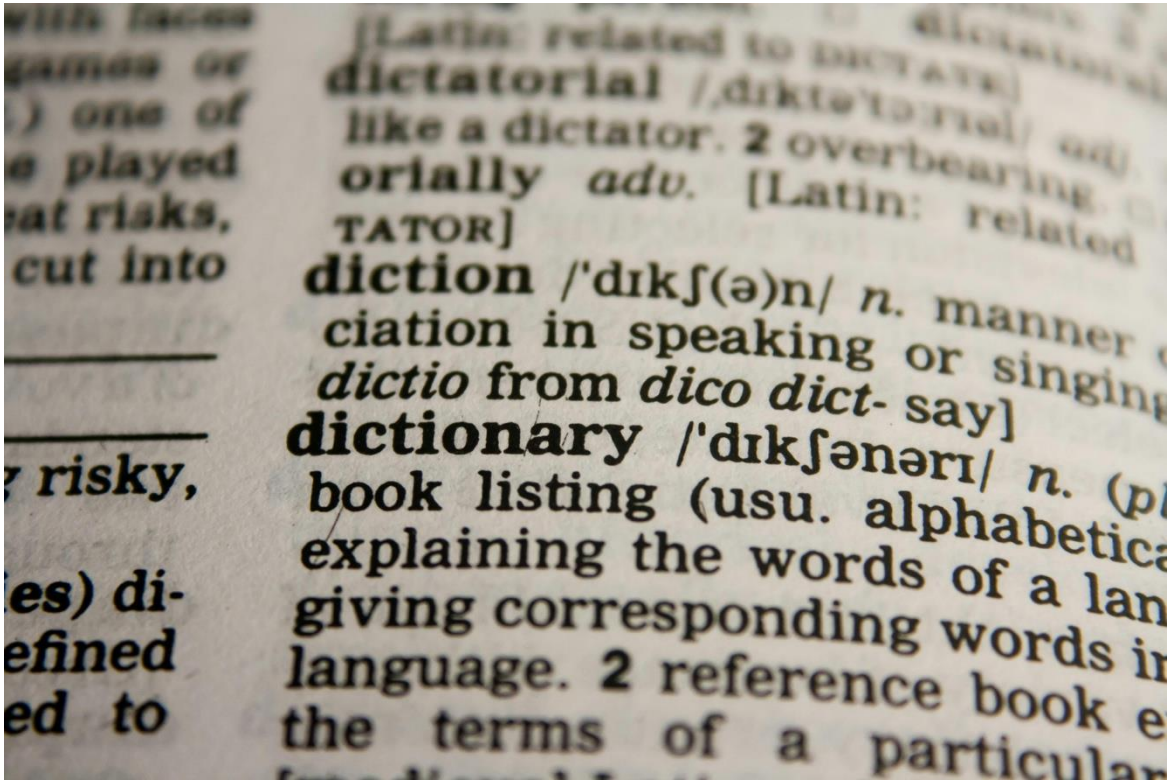
https://www.menti.com/al8qg1f2vuem

Rastak- Sornaye Nowruz

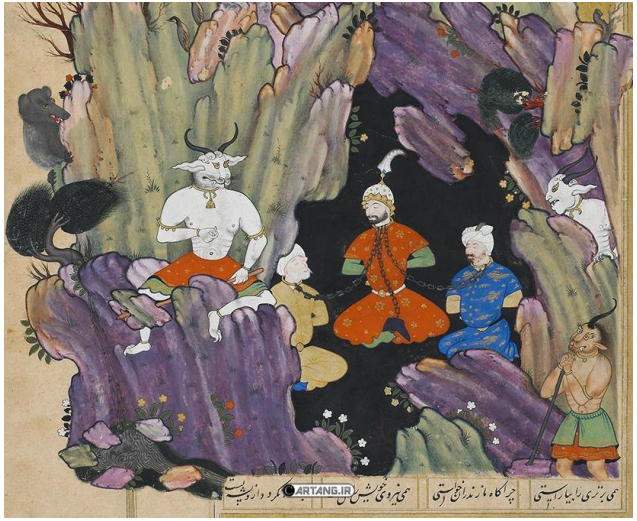
Break

https://www.youtube.com/watch?v=3IOMAXbwBI4

Dictionaries



How can we be better organised?



Dictionary in Python

- Dictionary in Python are used to store data values in key:value pairs.
- A dictionary is a collection which is ordered (in Python version 3.7), changeable and do not allow duplicates.
- Dictionary items are presented in key:value pairs and can be referred to by using the key name.
- Python dictionaries allow us to associate a value to a unique key, and then to quickly access this value.
- Dictionaries are changeable, meaning that we can change, add or remove items after the dictionary has been created.

Dictionary in Practice

- We are creating dictionary using curly braces or dict() method

```
marks_hw = {
    "0646": 89,
    "2134": 95,
    "2342": 93
}
# or
marks_exam = dict()
marks_exam["0646"] = 34
marks_exam["2134"] = 93
marks_exam["9134"] = 100
marks_exam["4134"] = 99
```

Can We Make Battles of Heroes More Organised?

Personal Dictionary

- Go to Ed on this link and create your own dictionary with any details you want. Show your results to the person next to you.

Python Containers (quick revision)

- Python collections can be list, tuples, dictionary
- Fundamental operations:
    - Initialize
    - Access
    - Modify elements (if possible)

Command Line Arguments

```
achi0646@AHH6CFCMWFJ wk04 % python3 battles.py rostam
```

Command Line Arguments

- Inputs are passed in through the command line instead of using input() function.

- Command line argument are stored in a variable (an array of string objects) called `argv` in module `sys`.

- First element will always be the script name.

Summary for part1

- Lists are contiguous blocks of memory.

- We can easily access (and manipulate) a group of related data.

- You can have lists of anything. As the programmer, you are deciding what memory you need to have and then how it is processed.

- You can mix different type of objects into the same list in Python but this may not work in the other languages. It is best that you use the *same type of objects* in your list.

Summary

- Another way of getting inputs from users – command line arguments which is stored in `sys.argv`.

- The first command line argument is always the script name.

- Dictionary items are presented in key:value pairs and can be referred to by using the key name.

Fully anonymous feedback form for COMP9001 students – Fill out form